

Question	Answer	Marks
4(a)	The (total) <u>kinetic</u> energy changes / decreases so (the collision is) inelastic	B1
	OR (relative) speed of approach not equal to / greater than (relative) speed of separation so (collision is) inelastic	(B1)
4(b)(i)	$p = mv$ or 0.25×3.6 or 0.25×5.2 $\Delta p = 0.25 \times (3.6 + 5.2)$	C1
	$= 2.2 \text{ kg m s}^{-1}$	A1
4(b)(ii)	$F = \Delta p / (\Delta)t$ $= 2.2 / 0.18$	C1
	$= 12 \text{ N}$	A1
	OR $F = ma$ and $a = (v-u) / t$ $= m\Delta v / t$ $= 0.25 \times (3.6 + 5.2) / 0.18$	(C1)
	$= 12 \text{ N}$	(A1)

Question	Answer	Marks
4(c)	$\frac{1}{2} mv^2 = mg(\Delta)h$ $\frac{1}{2} \times 0.25 \times 5.2^2 = 0.25 \times g \times h_1$ $h_1 = 5.2^2 / 2g$ $h_1 = 1.38$	C1
	$\frac{1}{2} \times 0.25 \times 3.6^2 = 0.25 \times g \times h_2$ $h_2 = 3.6^2 / 2g$ $h_2 = 0.66$ $h_2 / h_1 = 0.66 / 1.38$	C1
	ratio = 0.48	A1

Question	Answer	Marks